Editors' Suggestion

Neutrino Flavor Transformation in Neutron Star Mergers

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We present the first numerical relativity simulations including neutrino flavor transformations that could result from flavor instabilities, quantum many-body effects, or potential beyond standard model physics in neutron star mergers. We find that neutrino flavor transformations impact the composition and structure of the remnant, potentially leaving an imprint on the postmerger gravitational-wave signal. They also have a significant impact on the composition and nucleosynthesis yields of the ejecta.

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Introduction—Multimessenger astronomy is entering an exciting new era, with cutting-edge detectors set to observe gravitational waves (GWs) and their electromagnetic counterparts from compact object mergers at unprecedented rates [1–16]. This advancement promises to shed light on some of the most profound questions in astrophysics, particularly regarding the nature of dense matter in neutron stars [17–20]. While observations of kilonovae have confirmed that binary neutron star (BNS) mergers are a site of rapid neutron-capture (r-process) enrichment [21–24], the exact yields and detailed composition of the ejecta remain uncertain [25–29]. These uncertainties stem from the complex interplay of nuclear physics, neutrino interactions [30–37], and the hydrodynamical evolution of the merger remnant [38,39]. In particular, neutrino flavor oscillations, which alter the electron fraction (Y_e) of the outflows, play a crucial role in shaping the conditions for r-process element synthesis [40–45].

Accurate modeling of these complex phenomena requires not only solving general relativistic magnetohydrodynamics equations but also implementing a comprehensive scheme to evolve the neutrino radiation transport [46–50]. In numerical simulations, modeling neutrino transport typically requires solving a seven-dimensional Boltzmann equation to capture their evolution in both coordinate and phase space. However, this approach remains computationally infeasible due to its exorbitant cost [51,52]. As a result, alternative methods such as the nondeterministic Monte Carlo approach [53] and approximate schemes like truncated moment methods [54–60] are commonly employed [61,62].

So far, binary neutron star merger simulations have neglected neutrino flavor conversions. Based on the neutrino distributions from simulations without flavor transformation, the flavor evolution under neutrino-neutrino interactions [49,63–76] can occur on nanosecond timescales, which are orders of magnitude shorter than the timescales of hydrodynamic evolution, making them unfeasible to capture on the fly [77,78]. Important exceptions are the studies of postmerger accretion disks [42–44]. However, they considered only the evolution of disks around compact objects and did not consider the merger phase itself, which is the most dynamical phase of the binary evolution and that sets the disk initial structure and composition [79]. Additionally, there have been works on the effects of neutrino flavor conversions in the context of core-collapse supernovae [80–82], and substantial discussion regarding the outcomes of fast flavor conversions using either radiation-only simulations [83–90], or analyzing static snapshots from classical hydrodynamical simulations [91–95].

In this Letter, we consider, for the first time, the impact of rapid, lepton number-preserving neutrino flavor equilibration that could result from quantum many-body interactions [96], beyond standard model physics [97], or flavor instabilities [71] in neutron star mergers. We show that neutrino flavor transformation can impact the structure and composition of the remnant, possibly leaving detectable changes in the postmerger GW emission. Flavor transformation has a profound impact on the ejecta composition and associated nucleosynthesis yields, with relative isobaric abundances of some nuclear species changing by up to one order of magnitude.

Setup—We model neutrino flavor transformation using a Bhatnagar-Gross-Krook (BGK) collisional operator [98,99], $1/\tau_a(f^a - f)$, which models flavor conversion as a relaxation toward an asymptotic equilibrium state. In the previous equation, τ_a is the relaxation time, f is the neutrino distribution function, and f^a denotes f at its asymptotic mixing equilibrium state. Motivated by [96], we determine the mixing equilibrium states assuming detailed balance for neutrino many-body interactions; that is, the

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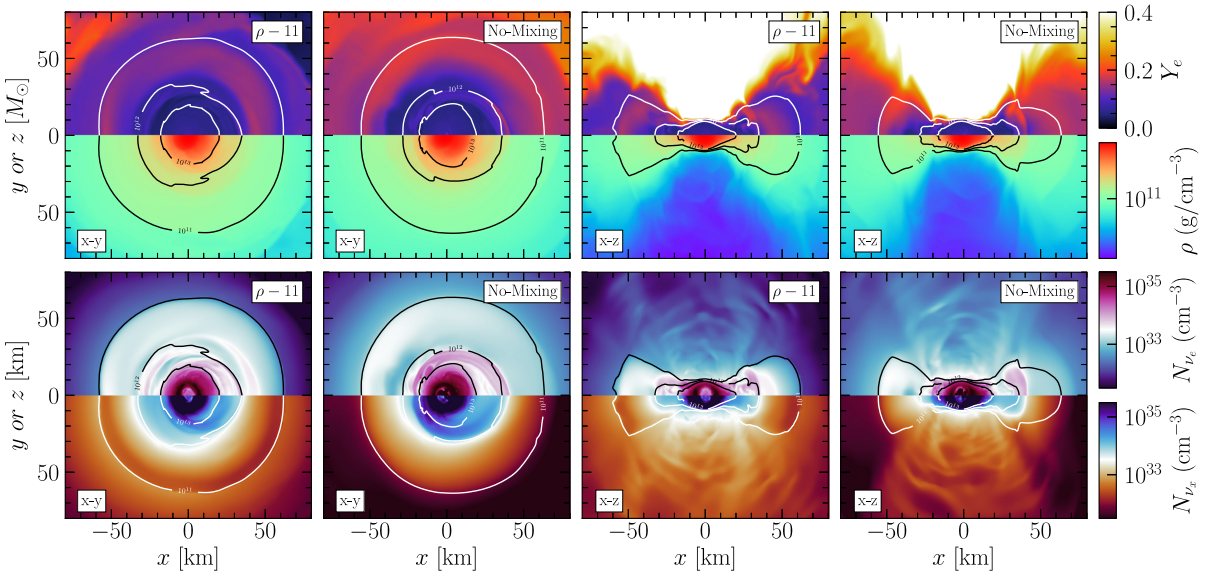


FIG. 1. Simulation snapshots in the equatorial x-y (meridional x-z) planes for the $\rho-11$ and the `No-Mixing` models at ∼15 ms after merger on the left (right) four panels, respectively. The top-row panels show electron fractions (upper half) and rest mass densities (lower half). The second-row panels display electron neutrino number densities (upper half) and heavy-lepton neutrino number densities (lower half). The contour lines denote where the density is at 10^{11}, 10^{12}, and 10^{13} g/cm^3. We find that after merger, the disk in the $\rho-11$ mixing model is more neutron rich than that of the `No-Mixing` model. Because of flavor conversion effects transforming electron (anti) neutrinos to heavy-lepton neutrinos, we also find that electron neutrinos are more abundant in the `No-Mixing` than the $\rho-11$, and vice versa for heavy-lepton neutrinos.

neutrino number densities are chosen such that the reactions $\nu_e\bar{\nu}_e \leftrightarrows \nu_\mu\bar{\nu}_\mu \leftrightarrows \nu_\tau\bar{\nu}_\tau$ are in equilibrium. Since only 4 neutrino species are considered, i.e., $\nu_e, \nu_x, \bar{\nu}_e, \bar{\nu}_x$, where x stands for μ and τ (anti)neutrinos combined, the assumption yields $4n_e^a n_{\bar{e}}^a = n_x^a n_{\bar{x}}^a$, where n is the number densities of neutrinos.

We perform BNS merger simulations with `THC_M1` [60,100–102]. Our code evolves the (lab frame) neutrino number density n, energy E, and flux F^α using conservation equations in the $3+1$ form (see in [60]), where on the right-hand side we add the BGK flavor conversion operators using the corresponding relaxation times and calculated equilibrium values [103]. For this work, we adopt the closure prescription and neutrino interaction rates described in Supplemental Material [104]. We do not consider neutrino-lepton inelastic scattering effects [105,106] in this work. The neutrino mixing operators are implemented in an operator-split fashion. Note that, in our model, the neutrino flavor decoherence effects cannot be fully captured, since we do not evolve the off-diagonal components of the neutrino density matrix. Nevertheless, this model has been shown to quantitatively capture the relaxation to equilibrium of the fast flavor instabilities in local simulations [99], and is adequate for the scenario we are considering here, in which detailed balance is quickly realized in flavor space.

We construct initial data assuming nonrotating equal-mass neutron stars with mass $1.35M_\odot$ and an initial separation of 45 km, corresponding to ∼3–4 orbits prior to merger, using the Lorene pseudospectral code [107]. Neutron star matter is described using the DD2 equation of state [108,109]. We employ 7 levels of adaptive mesh refinement [110,111] using the Carpet driver [112,113] of the Einstein Toolkit [114,115]. The innermost refinement level covers each star prior to merger, as well as the central part of the merger remnant. We present three sets of simulations, (i) no neutrino mixing (`No-Mixing`), (ii) neutrino mixing with density threshold 10^{11} g/cm^3 ($\rho-11$), and (iii) neutrino mixing with density threshold 10^{13} g/cm^3 ($\rho-13$). All simulations were conducted with both low resolution (LR) and standard resolution (SR), where the finest grids have spacingof $h = 0.167M_\odot \simeq 246$ m and $0.125M_\odot \simeq 184$ m, respectively. We discuss the SR results in the text. The LR results agree qualitatively; see Supplemental Material [104] for more information. Similar to [80,81], in the $\rho-11$ ($\rho-13$) simulations, we use uniform relaxation times $\tau_a = 0.4$ ns [116] for regions where the rest mass density ρ is below 10^{11} g/cm^3 (10^{13} g/cm^3), and infinite relaxation times when ρ is above the threshold. The $\rho-13$ model is chosen because the approximate decoupling densities of heavy-lepton neutrinos are 10^{13} g/cm^3. It is also roughly the density of the surface of the remnant [117], inside which the interaction with matter should dominate over the flavor conversion effects. The $\rho-11$ model considers a case in which flavor transformations become important only in the optically thin region, as electron-type neutrinos also decouple below density 10^{11} g/cm^3. Note that our simulations are subject to uncertainties due to finite-resolution effects [118] and the

use of moment-based scheme for neutrino transport [61]. However, the effects we report are larger than the estimated numerical uncertainties, which will be shown in the following text.

Results—Our simulations start from the late inspiral phase of the BNS, the stars merge ∼15 ms from the beginning of the simulation. Since DD2 is a stiff equation of state the remnants are stable against gravitational collapse. We evolve the system for an additional ∼30 ms postmerger to be able to capture the bulk of the dynamical mass ejection. In Fig. 1, we show the distribution of electron fractions, rest mass densities, electron neutrino, and heavy-lepton neutrino number densities on the x-y and x-z planes for the $\rho-11$ and the `No-Mixing` simulations. These snapshots, taken at ∼15 ms postmerger, capture the general dynamics of the remnant disk. In the equatorial plane and lower angle polar regions, we observe that the electron fraction of the accretion disk surrounding the merger remnant is substantially lower in the $\rho-11$ case than that in the `No-Mixing` simulation. In particular, Y_e inside the 10^{11} g/cm^3 density contours also seem to be lower in the $\rho-11$ case, possibly due to oscillations in the neutrino flavor conversion surface due to the spiral density waves [119]. Moreover, we observe mixing between the neutrino driven wind and the disk at intermediate latitudes, which might not be easily appreciated through the static snapshots in Fig. 1. In addition, as the electron (anti) neutrinos undergoing $\nu_e, \bar{\nu}_e \to \nu_x, \bar{\nu}_x$ flavor conversions, we find electron neutrinos to be less abundant in the outer disk (i.e., $\rho < 10^{11}$ g/cm^3) in the $\rho-11$ simulation than the `No-Mixing` simulation. Heavy-lepton neutrinos exhibit the opposite trend.

We compute the remnant binding energies in the upper panel of Fig. 2, by integrating the GW energy losses. The $\rho-13$ simulation has the highest GW luminosity and the most bound remnant, followed by the $\rho-11$ simulation, and then the least in the `No-Mixing` simulation. In particular, the binding energies for the $\rho-13$ remnants are ∼20% greater than the `No-Mixing` remnants. Consistently, we find that these remnants have higher maximum density and smaller central value of the lapse function, as shown in Supplemental Material [104]. This could be the result of the enhanced thermal and electron (anti)neutrino losses in the outer layers of these remnants favored by neutrino flavor transformation. We remark that a similar effect was recently reported by [97] in the context of core-collapse supernovae. This result suggests that gravitational wave observations could, at least in principle, constrain the physics of neutrino flavor transformation. We have also analyzed the postmerger GW spectra in the lower panel of Fig. 2 and found evidence of a shift in the peak frequency of the postmerger signal as a result of mixing, with the discrepancy being up to 100 Hz. However, we note here that such frequency shifts might be artificially exaggerated because of the double-peak feature seen in the

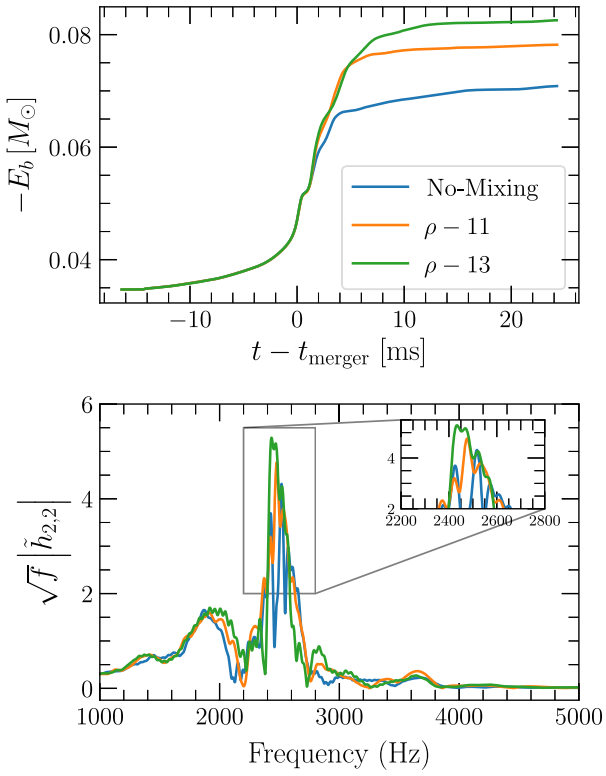


FIG. 2. Upper panel: evolution of the gravitational binding energy in the three models. The $\rho-13$ remnant is the most bound, followed by the $\rho-11$ simulation, and then by the `No-Mixing` simulation. Notably, the binding energy for the $\rho-13$ remnant is ∼20% greater than the `No-Mixing` remnant. Lower panel: GW spectra in the three models. The postmerger peak frequencies shift from high to lower in the `No-Mixing`, the $\rho-11$, and the $\rho-13$ simulations. The difference between the peak frequency is of ∼100 Hz. However, while the trend in binding energy is robust with respect to grid resolution, the frequency shifts are not, so we cannot exclude that they are a numerical artifact.

`No-Mixing` spectrum. Moreover, we could not resolve this trend in the lower-resolution simulations, so we cannot conclusively rule out that it is a numerical artifact.

To quantify the ejecta properties, we measure the flux of unbound material satisfying the geodesic criterion at a sphere located at $\sim 200 M_\odot$ (295 km) away from the center of the simulation domain. Figure 3 compares the electron fraction profiles and nucleosynthesis yields at ∼32 years after merger [120] for the three models. Because of the flavor conversions from electron neutrinos to heavy-lepton neutrinos, the induced deficits of electron neutrinos lead to a decrease in the neutrino reabsorption $\nu_e + n \to p + e^-$. Therefore, the mixing simulations have more neutron-rich ($Y_e < 0.2$) materials in the ejecta, with the $\rho-11$ simulation having the most, followed by $\rho-13$ simulation, and the `No-Mixing` simulation the least. The $\rho-11$ simulation generates up to 5 times more neutron-rich ejecta than the `No-Mixing` simulation, while $\rho-13$ simulation

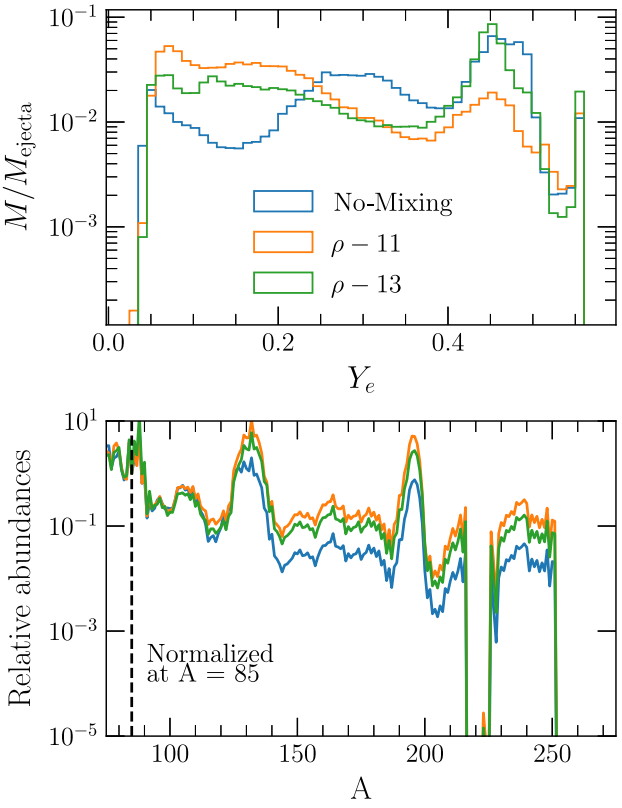


FIG. 3. Upper panel: histograms of the electron fraction distributions of the ejecta. Because of neutrino mixing, the ejecta are generally more neutron rich in the $\rho-11$ simulation and the $\rho-13$ simulation than that in the No-Mixing simulation. The $\rho-11$ simulation has the most low (<0.2) Y_e material in the ejecta, followed by $\rho-13$ simulation, and then by the No-Mixing simulation. Lower panel: the relative abundances of nuclei of mass number A formed in the ejecta. We normalize the yields at $A=85$. Note that the yields for lanthanides and heavy elements are one order of magnitude higher in the $\rho-11$ simulation than the No-Mixing simulation. Because the yields are determined by the r-process, which will be enhanced (suppressed) by more (less) neutron-rich ejecta, the ranking between the three models is consistent as the electron fraction histograms.

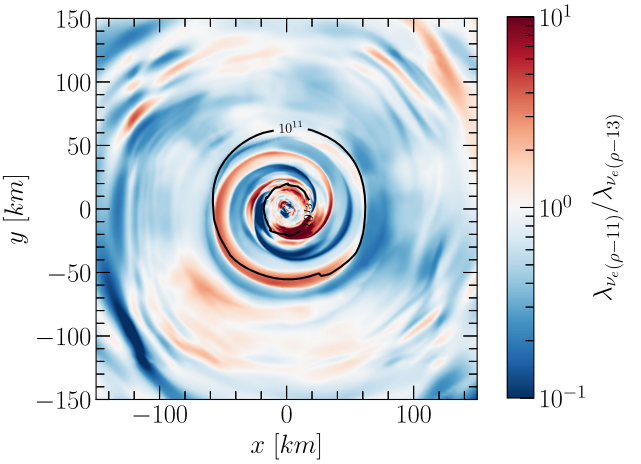


FIG. 4. Ratio of the electron neutrino absorption rates for the $\rho-11$ simulation over the $\rho-13$ simulation on the x-y plane, taken at ~15 ms after merger. The contours are drawn for rest mass density of 10^{11} and 10^{13} g/cm^3. We see greater electron neutrino absorption rates in the $\rho-13$ simulation than those of the $\rho-11$ simulation for the outer disk region where density is below 10^{11} g/cm^3.

produces up to twice as much. Both differences significantly exceed the reported 10%–30% numerical errors estimated from comparisons between M1 and Monte Carlo neutrino transport schemes [61,124]. The lower panel of Fig. 3 presents the corresponding nucleosynthesis yields, calculated using the convolution method as in our previous work [31,123]. Because of the strong dependence of r-process nucleosynthesis on the ejecta Y_e, we observe an order-of-magnitude difference in lanthanide and heavy element production between the $\rho-11$ and the No-Mixing simulations, and a factor of 2 between the $\rho-13$ and the No-Mixing simulations. Notably, the differences caused by flavor conversion effects range from approximately 200% to 1000%.

Furthermore, we find that changing the neutrino mixing conditions gives rise to quantitatively different ejecta properties. In the $\rho-13$ simulation, the mixing is turned on in the inner disk where density is between 10^{11} and 10^{13} g/cm^3. In this region, electron (anti)neutrinos remain in thermal equilibrium with the medium while undergoing flavor conversion to heavy-lepton neutrinos. The latter can freely propagate outward. Once these heavy-lepton neutrinos reach the outer disk where the density is below 10^{11} g/cm^3, a sufficient amount of them convert back into electron neutrinos, leading to an excess of electron neutrinos compared to the $\rho-11$ simulation. Therefore, we see in Fig. 3, the $\rho-13$ simulation yields up to 3 times less neutron-rich ejecta, and correspondingly fewer heavy elements than the $\rho-11$ simulation.

Such effects are further illustrated in Fig. 4, which shows the ratio of electron neutrino absorption rates λ_{ν_e} between the $\rho-11$ and the $\rho-13$ simulations. In neutron-rich ejecta, the electron neutrino absorption rate is approximately proportional to the electron fraction equilibration rate [10,11,21]:

$$\frac{dY_e}{dt} = \lambda_{\nu_e}(1-Y_e) - \lambda_{\bar{\nu}_e} Y_e \approx \lambda_{\nu_e}. \quad (1)$$

In the outer disk where density is below 10^{11} g/cm^3, the electron neutrino absorption rate is significantly higher in $\rho-13$ simulation than those of the $\rho-11$ simulation, leading to a more rapid increase in Y_e for the corresponding ejecta. This explains the less neutron-rich ejecta and heavy element produced in the $\rho-13$ simulation.

Conclusion—We report on the first numerical relativity simulations including the dynamical effects of lepton number-preserving neutrino flavor equilibration that could be driven by quantum many-body effects, beyond standard model physics, or flavor instabilities. Our study demonstrates that such transformations can impact the structure and composition of the merger remnant and of its accretion disk. At high densities, the conversion of electron-type neutrinos and antineutrinos into heavy-lepton neutrinos accelerates the contraction of the outer layers of the remnant. This is manifested by a significant increase, of up to ∼20%, in the overall postmerger GW luminosity, which could be probed with future GW experiments, such as Cosmic Explorer [125] and the Einstein Telescope [126]. At lower densities, the depletion of electron-type neutrinos due to flavor conversion results in more neutron-rich ejecta and boosts the production of heavy r-process elements with $A \gtrsim 120$ by up to one order of magnitude. We note that such differences are results of our neutrino flavor transformation treatments and choice of mixing prescriptions. In different flavor transformation scenarios, e.g., when considering the effects of the fast-flavor instability, the quantitative impact might change.

Interestingly, not only whether or not neutrino flavor transformation occurs but also where it occurs has a strong impact on the nucleosynthesis. In particular, we examine two density-dependent neutrino mixing models where neutrino flavor conversions are considered everywhere with density below either 10^{11} or 10^{13} g/cm^3. While both cases have significantly more neutron-rich ejecta and higher abundances of heavy r-process elements than the `No-Mixing` runs, there are important differences between them. In particular, if neutrinos undergo flavor transformation within the disk, e.g., at densities between 10^{11} and 10^{13} g/cm^3, this boosts the neutrino losses from the disk, because the disk is optically thin to heavy-lepton neutrinos. Some of the escaping neutrinos are then converted back to electron-type neutrinos and absorbed into the ejecta, decreasing the neutron richness and $\langle A \rangle$ of the ejecta.

Future work will extend this work by exploring mixing prescriptions motivated by the fast-flavor instability [82,127] and considering a broader range of equations of states and merger outcomes. We also aim to include muonic interactions [128,129], along with kernels for neutrino-lepton inelastic scatterings and pair processes [105,130], which are missing in this work and other published BNS simulations.

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Data availability—The data that support the findings of this Letter are not publicly available upon publication because it is not technically feasible and/or the cost of preparing, depositing, and hosting the data would be prohibitive within the terms of this research project. The data are available from the authors upon reasonable request.

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